WORKED SOLUTION

STEP III Specimen paper, Question 5

Pure • Counting Equation Roots

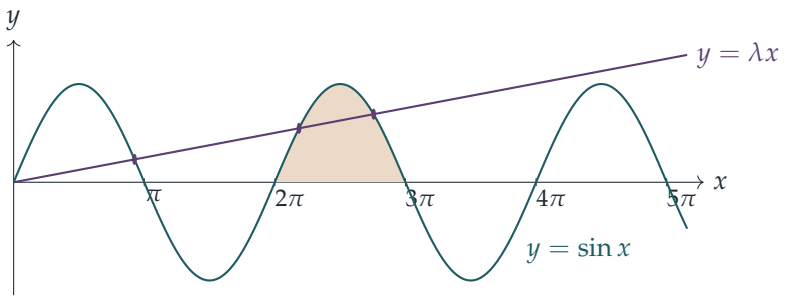
STEP III Specimen paper, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s3-q5

The key idea

Picture the line $y = \lambda x$ against the curve $y = \sin x$: solutions are crossings. The line can only meet the curve where $\sin x > 0$, that is on the positive "arches" $((2n-2)\pi, (2n-1)\pi)$, and because $\sin x$ is concave on each arch, the line crosses an arch either twice or not at all (apart from the first arch, which starts at the origin). For the small-ϵ estimate, the Maclaurin series $\sin x \approx x - \frac{1}{6}x^3$ does the work.



(Here $\lambda = 0.08$: one crossing on the first arch, two on the shaded second arch, and the line has risen above 1 before the third arch peaks, so $N = 3$.)

Solution

Part (i)

No solutions for $\lambda > 1$. The curve $y = \sin x$ has gradient $\cos x \leq 1$ and passes through the origin, and the line $y = x$ is its tangent there. For $x > 0$ we have $\sin x < x$ (the function $x - \sin x$ is zero at 0 and has derivative $1 - \cos x \geq 0$, equal to zero only at isolated points, so it is strictly increasing). Graphically, the curve lies strictly below its tangent $y = x$ for $x > 0$, and if $\lambda > 1$ the line $y = \lambda x$ lies above $y = x$ there. Hence

$$\sin x < x < \lambda x \qquad \text{for all } x > 0,$$

and there is no non-zero solution.

The case $\lambda = 1 - \epsilon^2$. For $x > 0$ the equation is equivalent to

$$\frac{\sin x}{x} = 1 - \epsilon^2.$$

First, a solution exists and it is the only one. On $(0, \pi]$ the function $\frac{\sin x}{x}$ is continuous, tends to 1 as $x \to 0^+$ and equals 0 at $x = \pi$, so it takes the value $1 - \epsilon^2$ somewhere in $(0, \pi)$; it is strictly decreasing there (the line through the origin to a point of the concave arch gets less steep as the point moves along it), so this happens just once. For $x \geq \pi$ there is nothing more: $\lambda x \geq (1 - \epsilon^2)\pi > 1 \geq \sin x$ because ϵ is small.

Now we locate it. Using $\sin x = x - \frac{1}{6}x^3 + \frac{1}{120}x^5 - \cdots$,

$$\frac{\sin x}{x} = 1 - \frac{x^2}{6} + \frac{x^4}{120} - \cdots = 1 - \epsilon^2 \quad \Longrightarrow \quad \frac{x^2}{6} - \frac{x^4}{120} + \cdots = \epsilon^2.$$

Ask yourself Why may we throw away the x^4 term? We don't yet know that x is small.

As $\epsilon \to 0$, $\frac{\sin x}{x} \to 1$, and since $\frac{\sin x}{x}$ is strictly decreasing from 1 on $(0, \pi)$, the root must tend to 0. So for small ϵ the root is small, and then x^4 is negligible beside x^2. To leading order $x^2/6 = \epsilon^2$, so

$$x \approx \epsilon\sqrt{6}.$$

(Keeping the next term, $x^2 \approx 6\epsilon^2 + \frac{6}{120}(6\epsilon^2)^2$, gives the refinement $x \approx \epsilon\sqrt{6}\left(1 + \frac{3}{20}\epsilon^2\right)$.)

Part (ii)

Let $g(x) = \sin x - \lambda x$, so the solutions are the zeros of g with $x > 0$. Since $\lambda x > 0$ for $x > 0$, every solution has $\sin x > 0$, so it lies on one of the arches

$$A_n = \big((2n-2)\pi,\ (2n-1)\pi\big), \qquad n = 1, 2, 3, \ldots$$

The first arch. At the origin the line has gradient $\lambda < 1$, smaller than the curve's, so the line starts below the curve; at $x = \pi$ it is above ($\lambda\pi > 0 = \sin\pi$). Since $\sin x$ is concave on A_1 and the line is straight, they cross exactly once there. So A_1 holds exactly one solution, which must be x_1, and $0 < x_1 < \pi$.

Later arches. On $A_{n+1} = (2n\pi, (2n+1)\pi)$ with $n \geq 1$, $g''(x) = -\sin x < 0$, so g is concave and has at most two zeros. At both ends of the arch $\sin x = 0 < \lambda x$, so $g < 0$ there. The maximum of g is where $g'(x) = \cos x - \lambda = 0$, at $c_n = 2n\pi + \arccos\lambda$, and $c_n < (2n + \frac{1}{2})\pi$ because $\arccos\lambda < \frac{\pi}{2}$ for $\lambda > 0$. Thus g increases on $(2n\pi, c_n)$ and decreases on $(c_n, (2n+1)\pi)$, so:

- if $g(c_n) < 0$ the line misses the arch;
- if $g(c_n) = 0$ the line touches it at one point;
- if $g(c_n) > 0$ there are exactly two solutions, one in $(2n\pi, c_n)$ and one in $(c_n, (2n+1)\pi)$.

Also, once the line misses an arch it misses all later ones: if $x \in A_{n+2}$ then $x - 2\pi \in A_{n+1}$, so $\sin x = \sin(x-2\pi) < \lambda(x-2\pi) < \lambda x$.

So the arches the line crosses are $A_1, A_2, \ldots$ up to some last one, contributing $1, 2, 2, \ldots$ solutions, possibly followed by a single tangency. Since $N = 2R+1$ is odd there is no tangency, and the line crosses exactly the arches $A_1, \ldots, A_{R+1}$. Listing the solutions in increasing order, arch A_{n+1} ($1 \le n \le R$) contains x_{2n} and x_{2n+1}, with

$$2n\pi < x_{2n} < c_n < (2n+\tfrac{1}{2})\pi, \qquad 2n\pi < x_{2n+1} < (2n+1)\pi.$$

The first gives the second line of the required inequalities. Writing x_{2n+1} as $x_{2(n+1)-1}$, the second, together with $0 < x_1 < \pi$, gives $(2m-2)\pi < x_{2m-1} < (2m-1)\pi$ for $m = 1, \ldots, R+1$. This is the result.

Estimating N. For small λ the line crosses many arches, so $R \ge 1$. At x_{2R} we have $\lambda x_{2R} = \sin x_{2R} \le 1$, and $x_{2R} > 2R\pi$ from above, so

$$2R\pi\lambda < 1.$$

On the other hand the line misses $A_{R+2} = \big((2R+2)\pi, (2R+3)\pi\big)$. At its midpoint $(2R+\frac{5}{2})\pi$ we have $\sin x = 1$, so we need $\lambda(2R+\frac{5}{2})\pi > 1$ (otherwise $g \ge 0$ there and the line would meet the arch). Combining,

$$\frac{1}{2\pi\lambda} - \frac{5}{4} < R < \frac{1}{2\pi\lambda}, \qquad \text{so} \qquad \frac{1}{\pi\lambda} - \frac{3}{2} < N < \frac{1}{\pi\lambda} + 1.$$

When λ is very small, $\frac{1}{\pi\lambda}$ is huge and the bounded error is negligible in comparison, so

$$N \approx \frac{1}{\pi\lambda}.$$

This makes sense: the line reaches height 1 at $x = 1/\lambda$, and before that point there are about $\frac{1}{2\pi\lambda}$ complete periods of $\sin x$, each contributing two crossings.

Checks

- With $\epsilon = 0.01$ ($\lambda = 0.9999$), bisection gives the root 0.0244953, against $\epsilon\sqrt{6} = 0.0244949$; the ratio is 1.000015, exactly the $1 + \frac{3}{20}\epsilon^2$ correction.
- For $\lambda = 0.001$ the computer finds $N = 317$ roots, against $\frac{1}{\pi\lambda} \approx 318.3$; for $\lambda = 0.01$ it finds 31 against 31.8. All roots found for several λ obey the interval inequalities of (ii), and N lies within the bounds above.
- No roots were found for $\lambda = 1.0001$, 1.1, 2 or 5.

Common slips

- Forgetting that x_1 lives on the first arch alone: the origin is a crossing too, which is why that arch contributes one non-zero solution rather than two.
- Placing x_{2n} in the first half of its arch without a reason. The point is that g only starts decreasing after $c_n = 2n\pi + \arccos\lambda$, which is before the peak of $\sin x$.

Beyond STEP

The approximation $x \approx \epsilon\sqrt{6}$ is the first step of an asymptotic expansion, and the argument that the root must be small before we truncate the series is the kind of care analysis courses insist on. Counting solutions by concavity, monotonicity and the intermediate value theorem is standard first-year analysis; see Abbott, *Understanding Analysis*. A related classic is $\tan x = x$, whose roots approach $(n + \frac{1}{2})\pi$ and which appears in heat conduction and vibration problems.

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